Title Slide

- Team Name:-QUANTRIX
- Problem Title & ID:-**Privacy-First On-Device AI Mobile Companion with Adaptive Behavior Learning**(SW-3)

SPECTRE

Problem Understanding

Current Situation

Most AI assistants like ChatGPT, Gemini, Siri, and Alexa rely heavily on cloud processing, meaning user conversations and personal data are sent to external servers.

Key Problems

- Privacy Risk – Personal conversations and sensitive information are processed on remote servers.
- No Transparency – Users cannot see what the AI learns about them or verify if data is truly deleted.
- Limited Personal Awareness – Current assistants react to queries but do not understand daily habits, routines, or digital wellbeing.
- No Addiction Protection – Existing systems do not monitor or reduce unhealthy smartphone usage patterns.
- False Trade-off – Users must choose between powerful AI with poor privacy or private AI with limited intelligence.

Core Challenge

Design a mobile AI companion that runs fully on-device, learns user behavior locally, protects privacy, promotes healthy usage, and gives users full control over their data.

SPECTRE

Solution Overview

Aegis is a fully on-device AI companion that protects privacy while helping users understand and improve their digital habits.

Three Core Roles

- Smart Assistant – Personalized AI that chats and remembers user preferences locally
- Behavior Coach – Analyzes phone usage patterns and daily routines
- Wellbeing Guardian – Detects unhealthy usage and provides gentle nudges

Key Capabilities

- Privacy by Design – All processing and storage happen on-device
- Transparency – Memory Vault lets users view, edit, or delete learned data
- Behavior Awareness – Learns app usage and routine patterns
- Addiction Safeguards – Detects excessive screen time and sends helpful nudges

The Breakthrough:

- Today you either get a smart AI that takes your data, or a private AI that forgets everything. Aegis eliminates that tradeoff — intelligent, private, and fully yours.

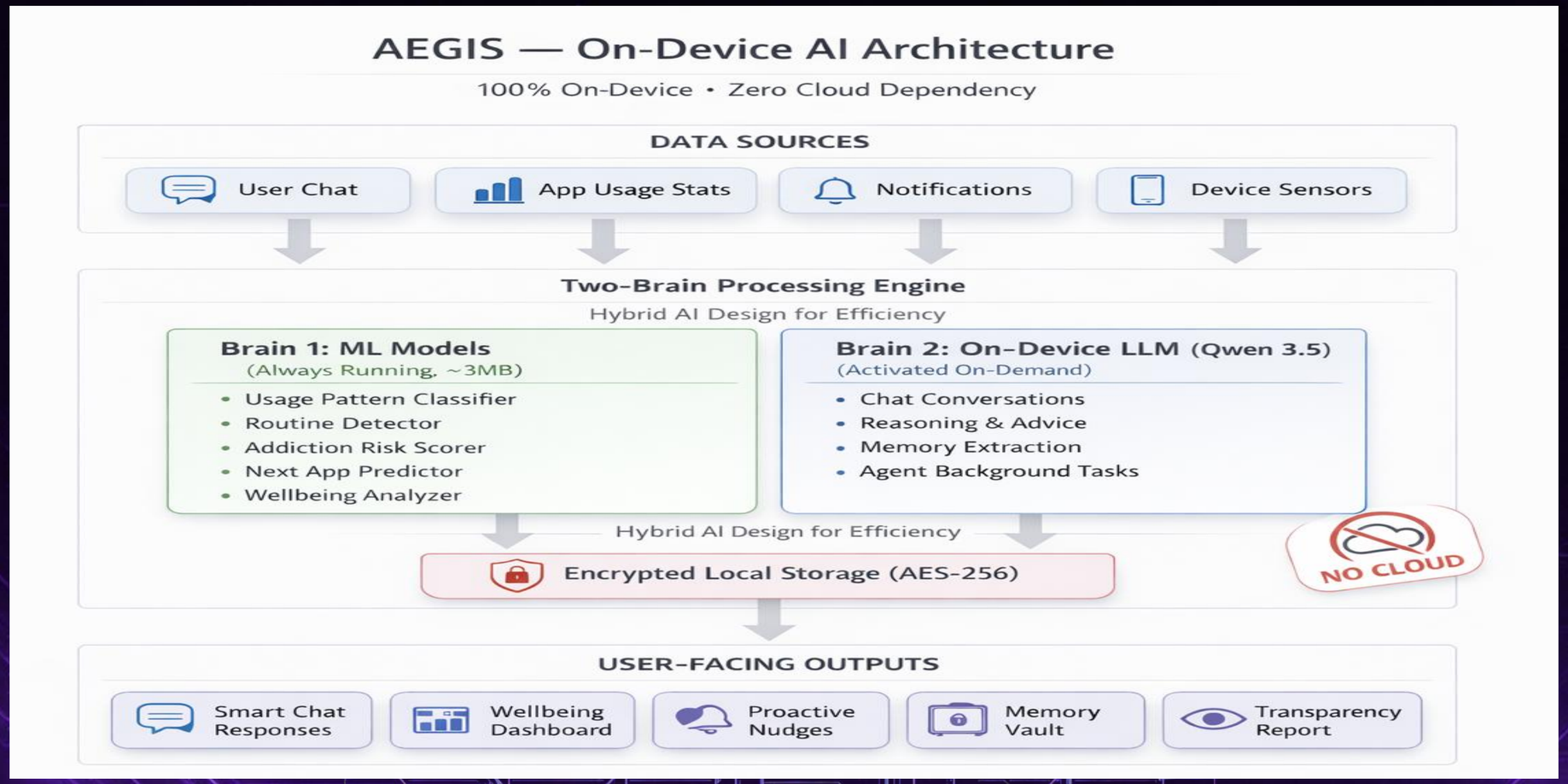
Tagline:

“Your AI. Your Phone. Your Rules.”

SPECTRE

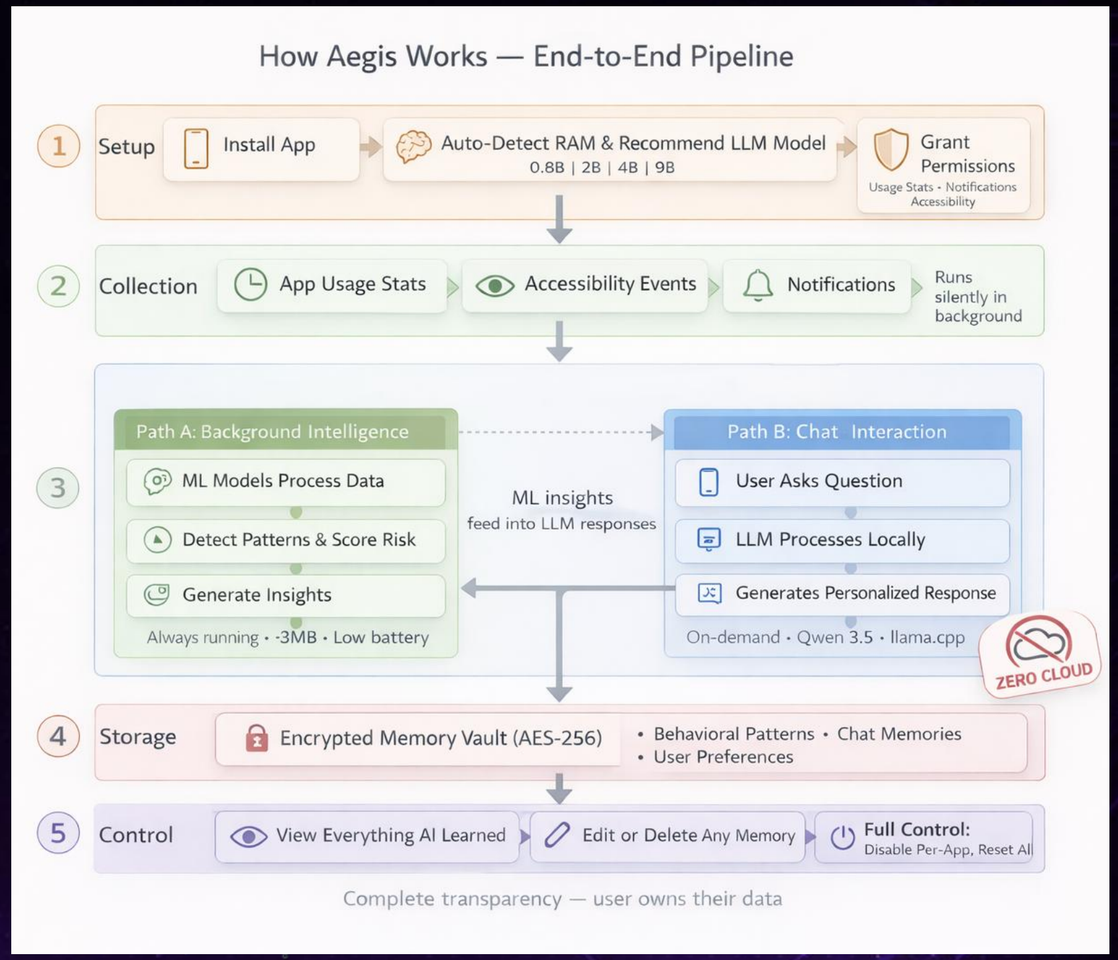
System Architecture / Block Diagram

"Everything runs on-device. Zero cloud dependency."



Methodology — How Aegis Works

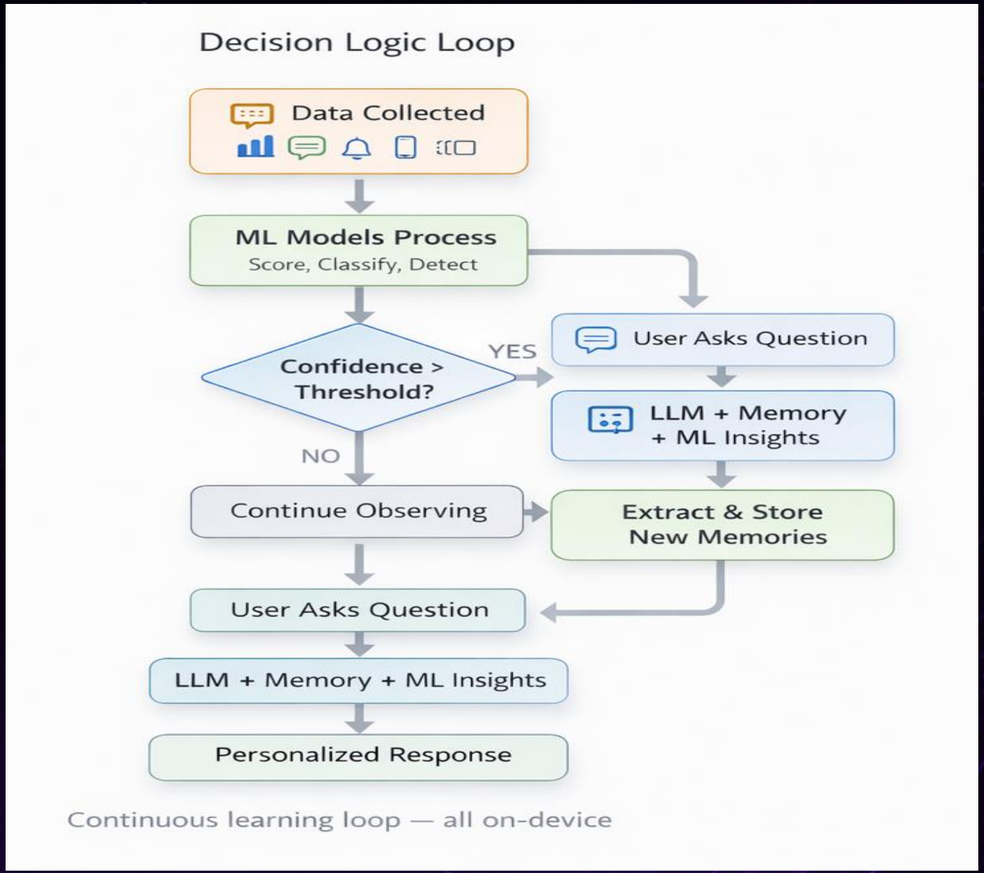
- **Step 1 – Setup:** App installs, detects device RAM, selects the optimal on-device LLM, and requests required permissions.
- **Step 2 – Data Collection:** Context signals are gathered from app usage stats, accessibility events, notifications, and device sensors.
- **Step 3 – Dual AI Processing:** Background ML models analyze behavior continuously, while the LLM activates on-demand for chat and reasoning.
- **Step 4 – Secure Storage:** Behavioral insights, preferences, and chat memories are stored locally with AES-256 encryption.
- **Step 5 – User Control:** Users can view, edit, delete AI memory, and control monitoring per app anytime.



SPECTRE

Algorithms & Logic

Component	Algorithm Used
Usage Classification	Decision Tree
Routine Detection	Hidden Markov Model
Addiction Scoring	Logistic Regression
App Prediction	Markov Chain
Wellbeing Trends	Moving Average (Time Series)
Memory Extraction	Regex + LLM Parsing
Memory Deduplication	Jaccard Similarity (> 0.7)
LLM Inference	llama.cpp + AirLLM



SPECTRE

Technology & Value Proposition

Layer	Tech
App	Flutter + Dart
State	Riverpod
LLM	Qwen 3.5 (GGUF)
Inference	llama.cpp + AirLLM
ML Models	Decision Tree, HMM, Logistic Regression
Storage	Hive (Encrypted)
Security	AES-256 + Android Keystore

What Makes Aegis Unique

- **Zero-Cloud AI** — All processing happens **100% on device**
- **Two-Brain System** — Background ML + On-Demand LLM
- **Adaptive Models** — Selects best model for device capability
- **Dual Inference** — Fast chat + deeper reasoning
- **Transparent Memory** — View, edit, or delete AI memories
- **Wellbeing Protection** — Detects unhealthy phone usage patterns

Aegis — Intelligent, Private, On-Device AI Companion

SPECTRE